

ABDALI, Kuwait

WMO: 405500

Lat: 30.067N

Lon: 47.683E

Elev: 25

StdP: 101.02

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	4.7	-10.6	1.5	13.2	-8.3	1.9	14.0	8.9	12.1	7.4	14.2	2.7	320	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	48.5	21.6	47.5	21.5	46.4	21.2	27.5	35.0	25.8	35.3	24.2	36.5	4.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	20.9	32.1	23.2	18.1	30.9	20.9	15.6	30.3	88.1	33.9	79.9	35.5	73.2	35.5	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.0	6.8	-0.2	50.5	1.9	0.8	-1.6	51.2	-2.7	51.6	-3.8	52.1	-5.2	52.7		
			-2.3	29.2	2.2	1.2	-3.9	30.1	-5.2	30.8	-6.5	31.4	-8.1	32.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	12.8	15.7	20.7	26.3	33.1	37.4	39.0	38.5	34.7	28.9	19.9	14.8	(6)
	DBStd	9.80	3.20	3.37	3.48	3.29	2.90	2.16	1.65	1.64	2.16	3.04	3.55	3.00	(7)
	HDD10.0	17	12	3	0	0	0	0	0	0	0	0	0	3	(8)
	HDD18.3	406	171	83	14	0	0	0	0	0	0	0	23	115	(9)
	CDD10.0	6179	100	163	330	488	716	822	901	883	742	587	296	150	(10)
	CDD18.3	3526	1	10	87	238	457	572	642	625	492	329	69	4	(11)
	CDH23.3	56765	4	84	883	2834	7124	10128	11686	11206	8026	4162	596	31	(12)
	CDH26.7	40728	0	13	329	1505	4860	7734	9206	8727	5721	2434	197	2	(13)
(14) Wind	WSAvg	2.9	2.4	2.8	2.9	2.9	3.0	4.0	4.0	3.2	2.9	2.4	2.2	2.4	(14)
(15) Precipitation	PrecAvg	315	67	42	47	29	7	1	1	0	1	12	45	63	(15)
	PrecMax	453	164	101	149	102	47	3	2	1	7	56	192	173	(16)
	PrecMin	170	12	1	5	1	0	0	0	0	0	0	1	5	(17)
	PrecStd	82	37	25	33	23	9	1	1	0	1	15	46	44	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	28.4	37.6	40.6	46.0	49.2	50.1	49.2	46.8	42.2	33.6	26.3	(19)
		MCWB	13.7	15.1	16.7	18.7	20.5	21.0	21.6	22.1	21.5	20.3	17.2	14.0	(20)
	2%	DB	21.9	25.8	32.7	37.4	43.9	47.6	48.8	48.2	44.8	39.9	31.0	24.0	(21)
		MCWB	13.4	14.3	16.0	17.9	20.5	21.2	21.4	21.8	21.4	19.8	16.8	14.8	(22)
	5%	DB	20.4	23.9	30.0	35.7	42.3	46.4	47.6	47.2	43.5	38.4	28.9	22.4	(23)
		MCWB	13.1	13.7	15.2	17.3	19.7	20.7	21.5	21.6	20.8	19.5	16.7	14.4	(24)
	10%	DB	18.9	22.2	27.8	33.8	40.7	45.1	46.5	46.0	42.3	36.6	26.8	20.8	(25)
		MCWB	12.5	13.0	14.8	17.0	19.3	20.3	21.2	21.6	20.3	19.3	16.1	14.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	18.3	20.0	21.6	23.8	26.3	27.6	29.5	28.4	27.3	23.1	19.5	(27)
		MCDB	20.0	22.1	26.6	28.9	34.1	36.6	35.3	35.0	32.7	33.2	26.7	21.7	(28)
	2%	WB	16.2	17.0	19.0	20.4	22.2	22.7	25.3	28.2	26.7	25.2	20.8	18.1	(29)
		MCDB	18.9	21.9	25.1	29.9	36.0	44.3	38.1	35.2	34.8	29.5	25.6	20.7	(30)
	5%	WB	15.2	15.7	18.0	19.5	21.4	21.7	23.3	26.2	24.4	24.0	19.3	16.6	(31)
		MCDB	18.2	20.3	24.6	29.8	35.8	44.8	42.3	37.6	37.1	29.7	24.8	19.7	(32)
	10%	WB	13.8	14.5	16.8	18.7	20.6	20.9	22.3	24.0	22.4	22.6	18.0	15.1	(33)
		MCDB	17.5	19.6	24.3	28.9	35.5	43.7	44.0	40.5	39.2	29.5	23.2	18.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	13.7	13.8	14.1	15.0	15.2	16.0	16.1	14.5	12.7	12.1	(35)
		MCDBR	12.7	13.9	17.0	15.9	15.6	16.2	16.4	16.8	16.7	16.3	14.7	13.4	(36)
	5% WB	MCWBR	6.2	5.6	5.9	5.3	5.2	5.4	5.8	5.7	5.8	6.3	6.5	6.4	(37)
		MCWBR	10.4	10.5	12.0	12.7	14.0	15.6	15.9	15.3	14.6	12.9	10.9	9.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.500	0.542	0.600	0.692	0.690	0.632	0.666	0.615	0.560	0.579	0.539	0.503	(40)
		taud	1.814	1.726	1.604	1.490	1.495	1.563	1.530	1.619	1.715	1.690	1.761	1.818	(41)
	Ebn at Noon	Ebn at Noon	695	708	701	658	664	701	675	704	724	666	650	664	(42)
		Edh at Noon	179	213	256	297	297	276	285	258	226	217	187	170	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	3.46	4.38	5.55	6.09	7.04	7.93	7.55	7.13	6.29	4.90	3.65	3.20	(44)
	RadStd	RadStd	0.24	0.26	0.22	0.32	0.33	0.23	0.21	0.17	0.20	0.21	0.27	0.29	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.29	N/A	N/A	+0.44	N/A	N/A	N/A	+85	+83

Nomenclature: See separate page